

Physical Chemistry (I)

Lecture 05

Real Gas

First Law of Thermodynamics



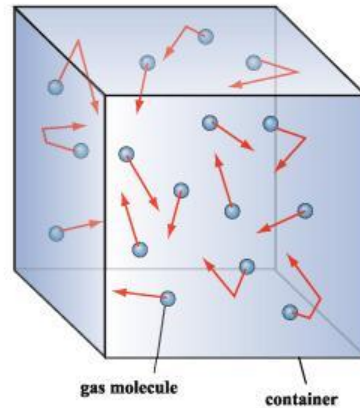
In the Last Class

- Kinetic molecular theory
 - Pressure: perfect gas law
 - Internal energy
- Equipartition Theorem
- Real gas
 - Compression factor
 - Virial expansion



Last class

Pressure



momentum change $\rightarrow$ force $\rightarrow$ pressure

$$pV = \frac{1}{3}nM\langle v^2 \rangle$$

$$pV = nRT$$



(Image: http://web2.uwindsor.ca/courses/physics/high_schools/2005/Brownian_motion/kinetictheory.jpg)

Last class

Internal Energy, U

“The total energy of a system”

For a perfect gas,

$$U = \sum_{i=1}^N \frac{1}{2} m v_i^2 = \frac{1}{2} M n \langle v^2 \rangle$$

$$U = \frac{3}{2} n R T$$



Equipartition Theorem

For a sample at thermal equilibrium,
the average value of each quadratic contribution
to the internal energy is $(1/2)k_B T$.

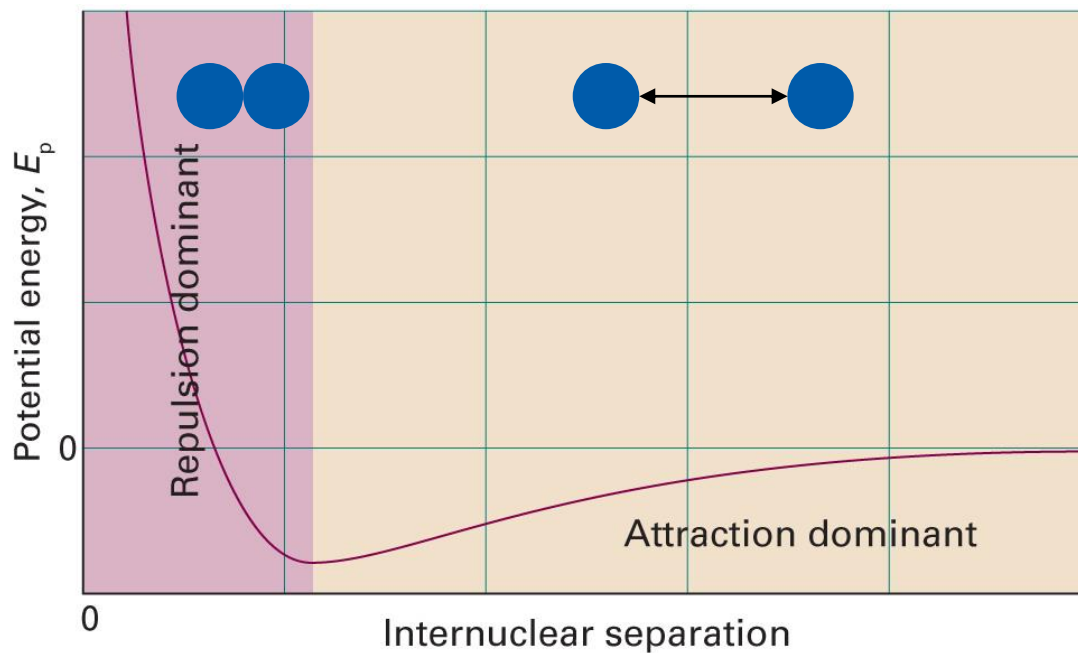
(Quadratic contribution: $E \propto p_x^2$ or $E \propto x^2$)

Translation + Rotation + Vibration

Last class

Real Gas

Now the molecules interact!



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1C.1)

Compression Factor Z

$$Z = V_m/V_m^0$$

Molar volume of a gas $V_m = V/n$

Molar volume of a perfect gas V_m^0

- $Z > 1$: $V_m > V_m^0 \rightarrow$ repulsion is dominant
- $Z \approx 1$: $V_m \approx V_m^0 \rightarrow$ perfect-gas-like behavior
- $Z < 1$: $V_m < V_m^0 \rightarrow$ attraction is dominant

Virial Expansion

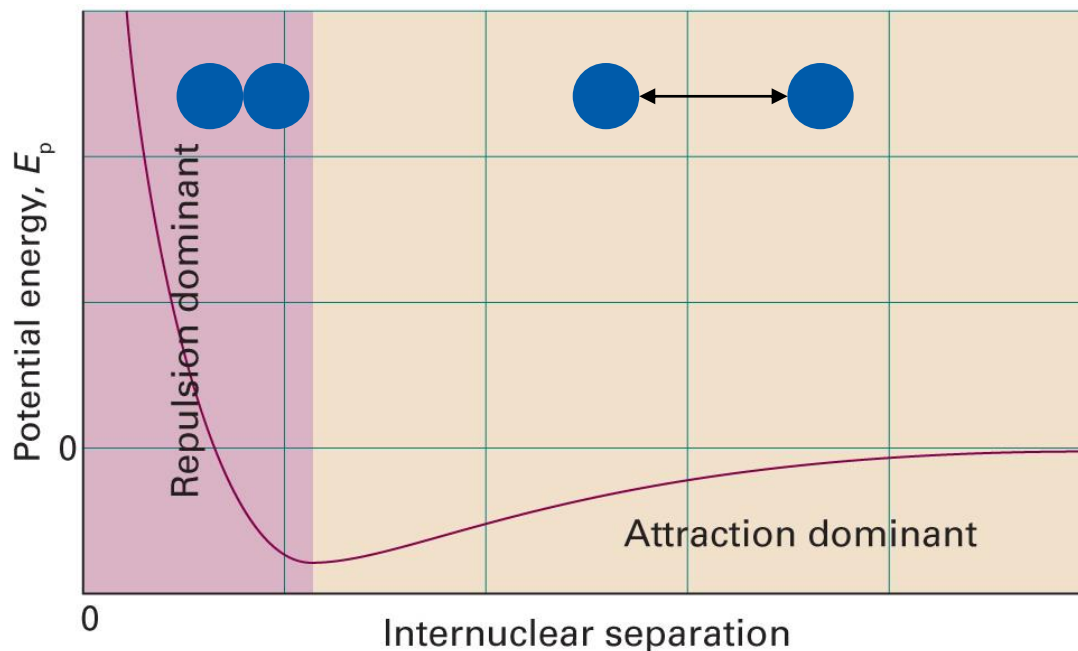
$$Z(p) = 1 + B'p + C'p^2 + \dots$$

$$Z(V_m) = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots$$

- Second virial coefficient B
 - $B > 0 \rightarrow Z > 1 \rightarrow$ repulsion is dominant
 - $B = 0 \rightarrow Z \approx 1$ (perfect-gas-like) for a larger region
 - Boyle temperature T_B : temperature where $B = 0$
 - $B < 0 \rightarrow Z < 1 \rightarrow$ attraction is dominant

Corrected Perfect Gas Law

Consider the effects of **repulsion** and **attraction**



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 1C.1)

van der Waals Equation

- **Repulsion:** decrease of the total volume

$$V \rightarrow V - nb$$

$$p = \frac{nRT}{V - nb}$$

- **Attraction:** decrease of the pressure

$p \sim$ (frequency of collisions) $\times$ (force of each particle)

decreases as $n/V \uparrow$

decreases as $n/V \uparrow$

$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$$

van der Waals Coefficients

$$p = \frac{nRT}{V - nb} - \frac{a n^2}{V^2}$$

- a : strength of attractive interactions
- b : strength of repulsive interactions
- Characteristics of each gas
- Independent of the temperature

Experimental Data

	$a / (\text{atm L}^2 \text{ mol}^{-2})$	$b / (10^{-2} \text{ L mol}^{-1})$
Ar	1.337	3.20
CO ₂	3.610	4.29
He	0.0341	2.38
Xe	4.137	5.16

Equation Manipulations

$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$$

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

can be represented by a **cubic equation**

van der Waals and Virial

- van der Waals equation

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

- Virial expansion

$$pV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

Can we connect the two?

Some Math

Target

$$pV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

$$\text{vdW: } p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

$$p = \frac{1}{V_m} \left(\frac{RT}{1 - b/V_m} \right) - \frac{a}{V_m^2}$$

$$pV_m = \frac{RT}{1 - b/V_m} - \frac{a}{V_m}$$

$$pV_m = RT \left(\frac{1}{1 - b/V_m} - \frac{a/RT}{V_m} \right)$$

Some More Math

Target

$$pV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

$$pV_m = RT \left(\frac{1}{1 - b/V_m} - \frac{a/RT}{V_m} \right)$$

As $b/V_m \ll 1$, we use

$$\frac{1}{1 - x} = 1 + x + x^2 + \dots \text{ (when } x < 1 \text{)}$$

$$pV_m = RT \left[\left\{ 1 + \frac{b}{V_m} + \left(\frac{b}{V_m} \right)^2 + \dots \right\} - \frac{a/RT}{V_m} \right]$$

Finally

Target

$$pV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

$$pV_m = RT \left[\left\{ 1 + \frac{b}{V_m} + \left(\frac{b}{V_m} \right)^2 + \dots \right\} - \frac{a/RT}{V_m} \right]$$

$$pV_m = RT \left[1 + \left(b - \frac{a}{RT} \right) \frac{1}{V_m} + \frac{b^2}{V_m^2} + \dots \right]$$

Hence,

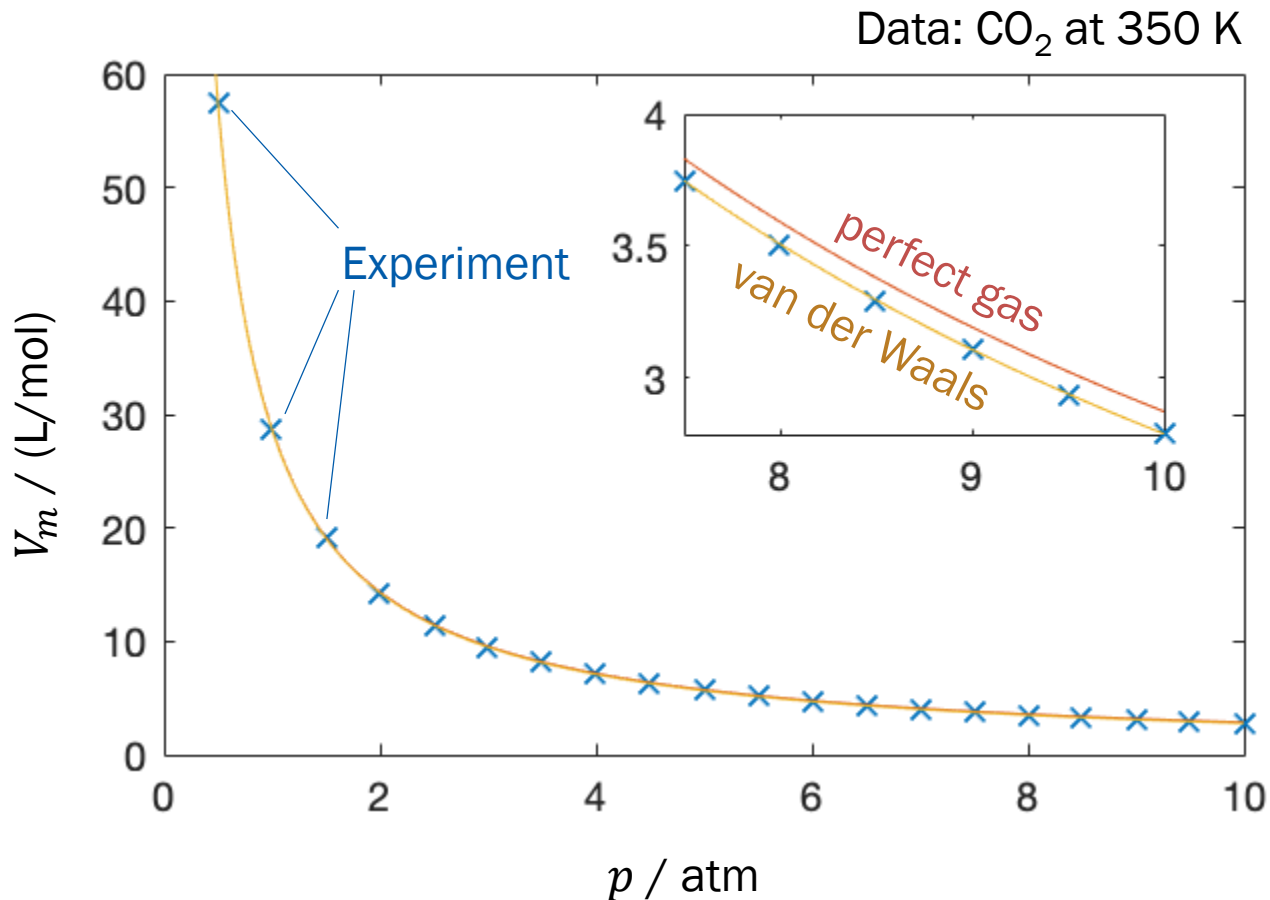
$$B = b - a/RT \text{ and } C = b^2$$



Also, from $B = b - a/RT = 0$, $T_B = a/(bR)$

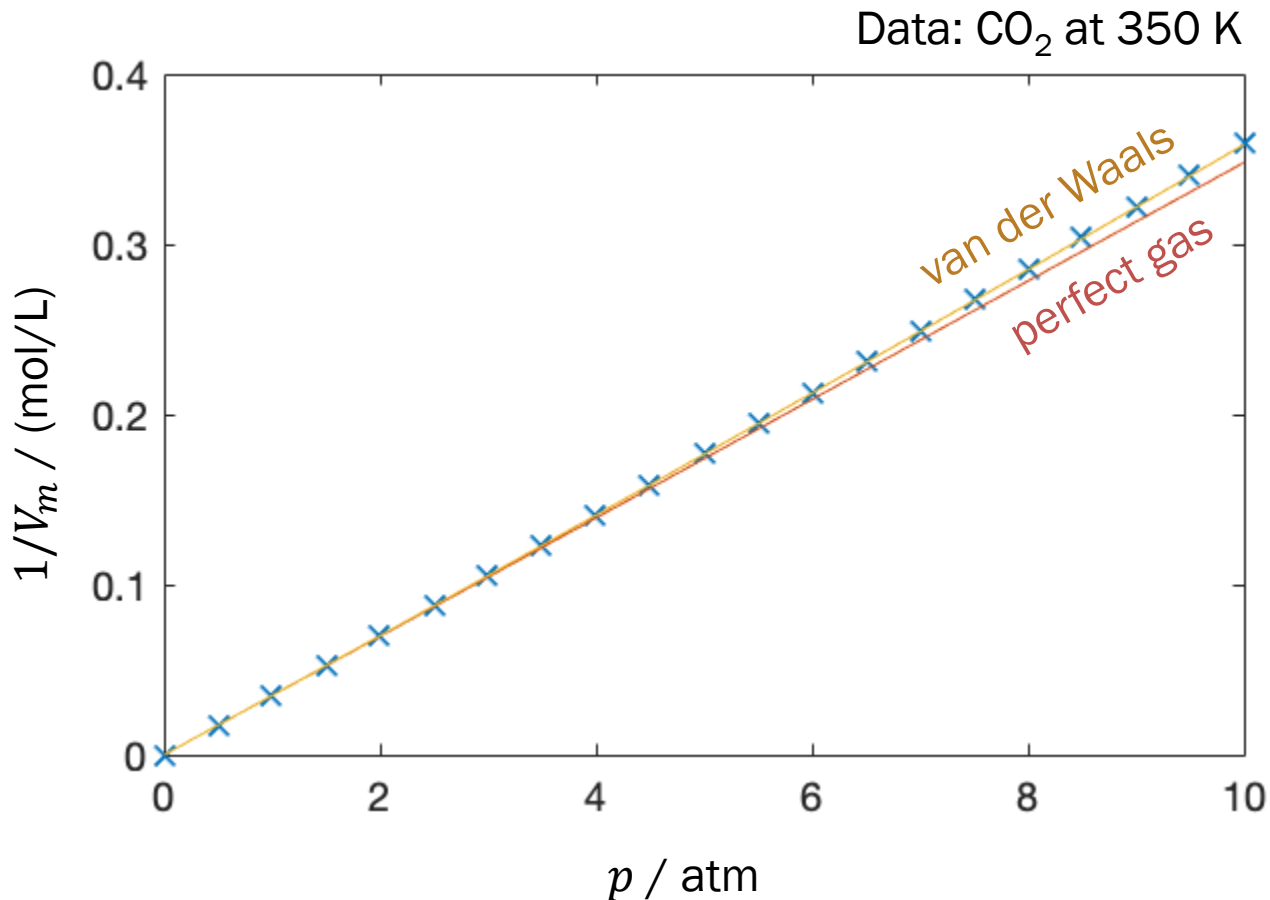
Not in Atkins

Validity of van der Waals



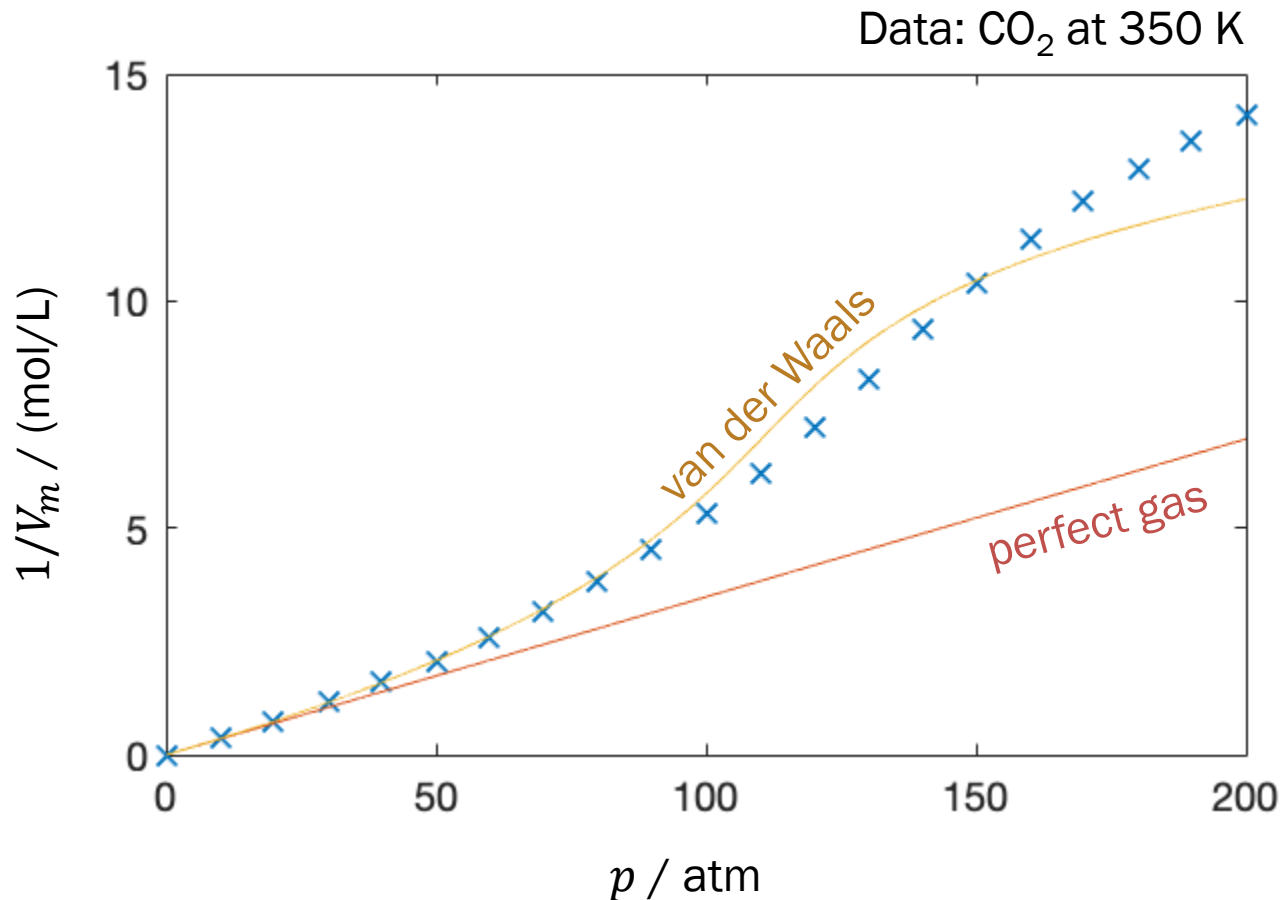
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Validity of van der Waals



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Is van der Waals Perfect?



Other Real Gas Equations

- van der Waals equation (1873)

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

- Modification of the attraction term
 - Redlich-Kwong equation (1949)

$$p = \frac{RT}{V_m - B_{RK}} - \frac{A_{RK}}{V_m(V_m + B_{RK})T^{1/2}}$$

- Peng-Robinson equation (1976)

$$p = \frac{RT}{V_m - B_{PR}} - \frac{A_{PR}}{V_m(V_m + B_{PR}) + B_{PR}(V_m - B_{PR})}$$

Other Real Gas Equations

- van der Waals equation (1873)

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

- Modification of the repulsion term
 - Guggenheim equation (1965)

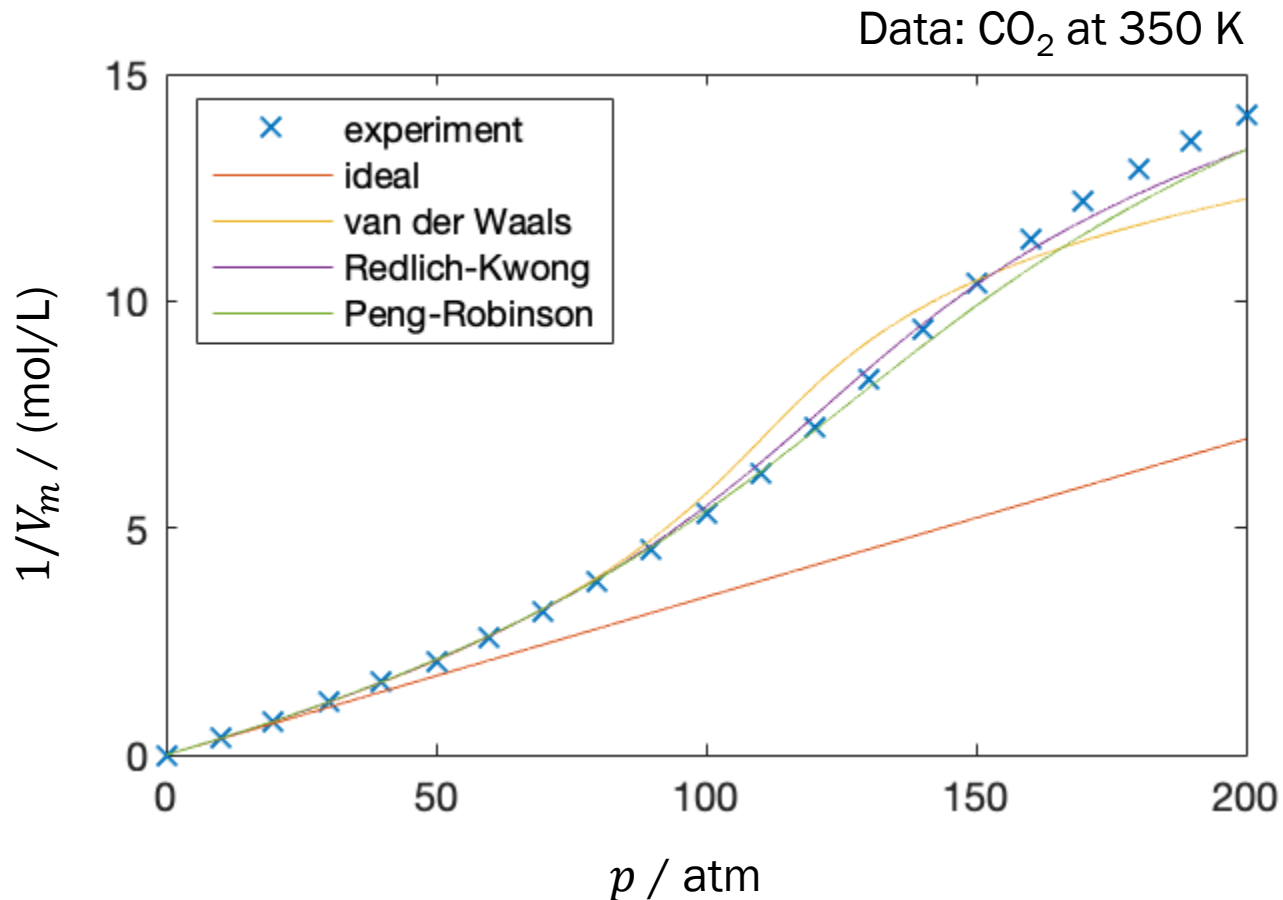
$$p = \frac{RT}{V_m(1 - y)^4} - \frac{A_G}{V_m^2}, \quad y = \frac{B_G}{V_m}$$

- Carnahan-Starling-van der Waals equation (1972)

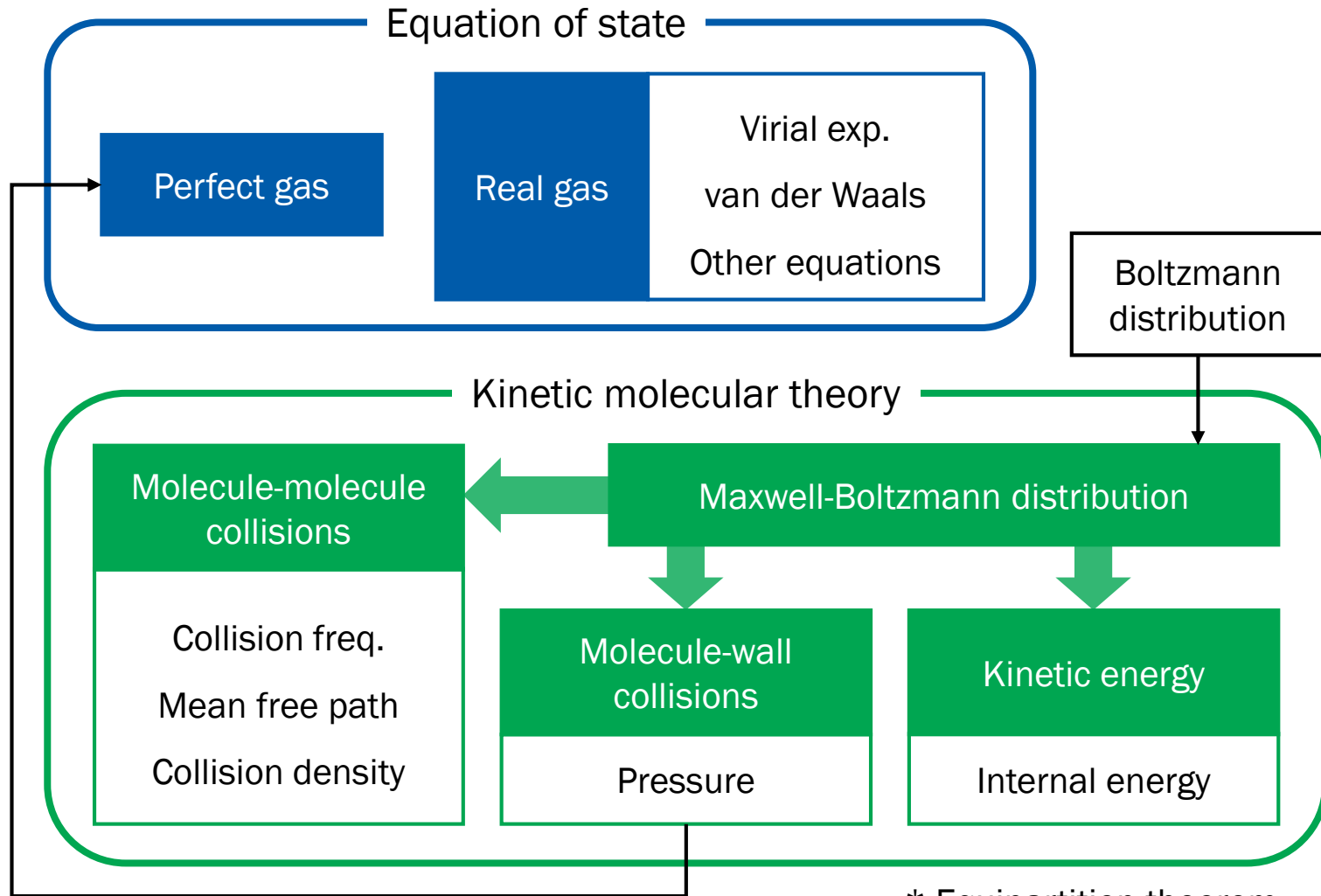
$$p = \frac{RT(1 + y + y^2 - y^3)}{V_m(1 - y)^3} - \frac{A_{CSW}}{V_m^2}, \quad y = \frac{B_{CSW}}{V_m}$$

Not in Atkins

Better Descriptions



Summary



* Equipartition theorem

3. The First Law



Chapter Overview

- First law of thermodynamics
 - Work and heat
 - Differential form of internal energy
- Enthalpy
 - Definition and properties of enthalpy
 - Differential form
 - Adiabatic change
- Thermochemistry



Internal Energy, U

“The total energy of a system”

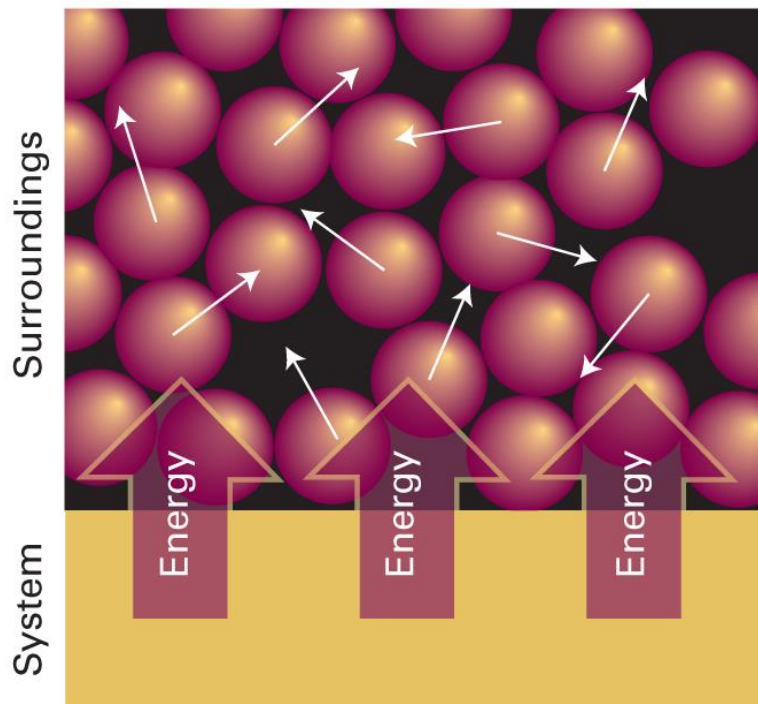
- Perfect gas (monatomic): $U = (3/2)nRT$
- Molar internal energy

$$U_m = U/n$$

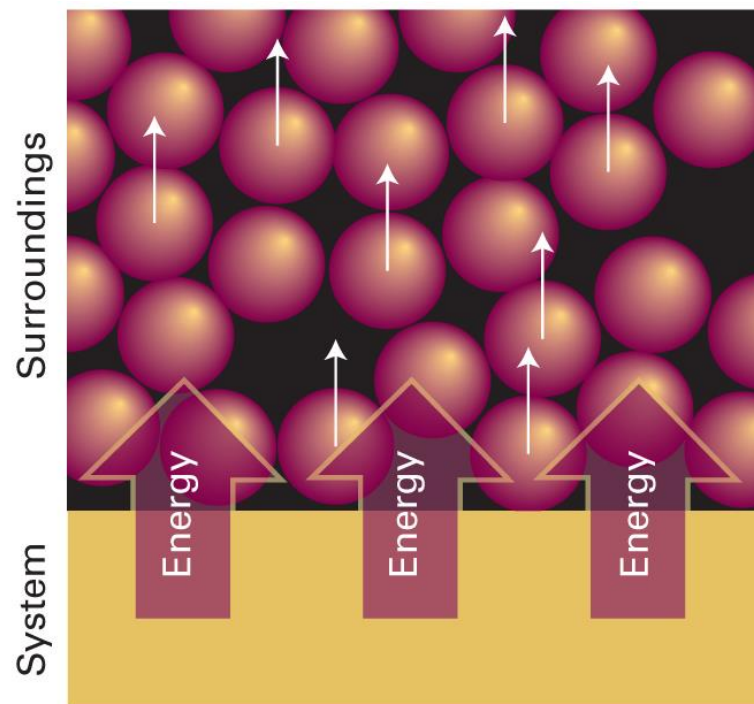
For a perfect gas, $U_m = (3/2)RT$

How can we change the internal energy?

Two Possible Ways



“Heat”



“Work”

First Law of Thermodynamics

“The internal energy of an isolated system is constant.”

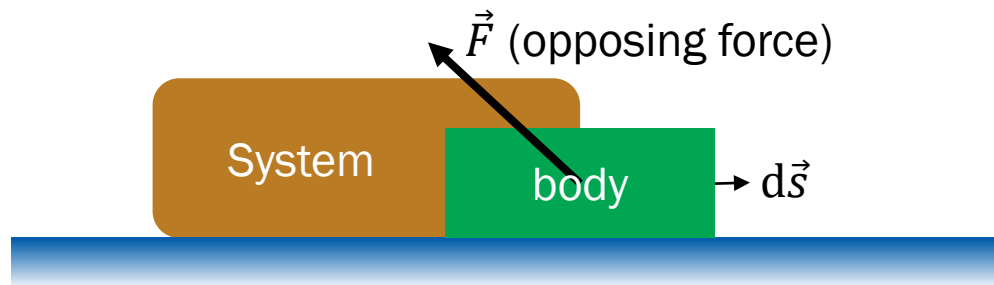
- Heat and work are equivalent ways of changing the internal energy of a system

$$\Delta U = q + w$$

$$dU = \delta q + \delta w$$

↑ ↑ ↑
small change of U small q small w

Work



- Work done on the body

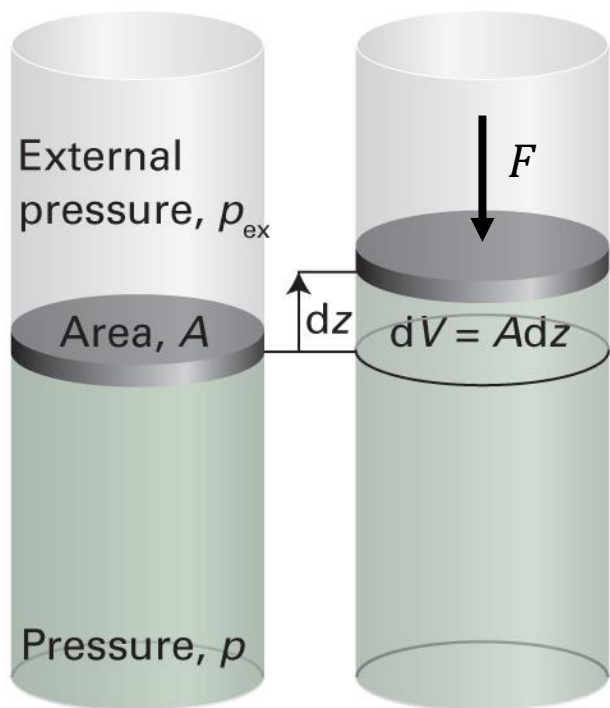
$$\delta w_{\text{body}} = -\vec{F} \cdot d\vec{s}$$

- Work done on the system

$$\delta w = \vec{F} \cdot d\vec{s}$$

where $\vec{F} \cdot d\vec{s} = F_x dx + F_y dy + F_z dz$

Expansion Work



$$\delta w = \vec{F} \cdot d\vec{s} = F_x dx + F_y dy + F_z dz$$

$$\text{As } dx = dy = 0,$$

$$\delta w = -|F|dz$$

$$\text{Since } p_{\text{ex}} = |F|/A,$$

$$\delta w = -p_{\text{ex}}Adz = -p_{\text{ex}}dV$$

Other Types of Work

$$\delta w = (\text{intensive factor}) \times (\text{extensive factor})$$

Type of work	dw	Comments	Units [†]
Expansion	$-p_{\text{ex}}dV$	p_{ex} is the external pressure	Pa
		dV is the change in volume	m^3
Surface expansion	$\gamma d\sigma$	γ is the surface tension	N m^{-1}
		$d\sigma$ is the change in area	m^2
Extension	$f dl$	f is the tension	N
		dl is the change in length	m
Electrical	ϕdQ	ϕ is the electric potential	V
		dQ is the change in charge	C
	$Q d\phi$	$d\phi$ is the potential difference	V
		Q is the charge transferred	C



Calculation of Expansion Work

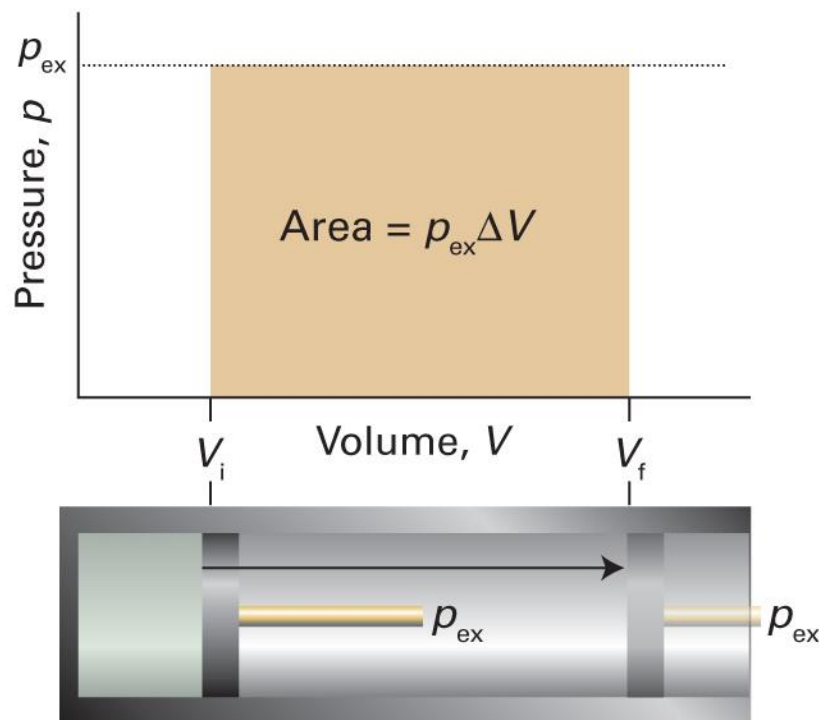
$$\delta w = -p_{\text{ex}} dV$$

- Total work for a change from **state i** to **state f**

$$w = \int_{\text{state } i}^{\text{state } f} \delta w = \int_{\text{state } i}^{\text{state } f} (-p_{\text{ex}} dV) = - \int_{V_i}^{V_f} p_{\text{ex}} dV$$

1. Expansion against constant external pressure
2. Reversible expansion

Constant External Pressure



$$w = - \int_{V_i}^{V_f} p_{\text{ex}} dV$$

$$w = -p_{\text{ex}} \int_{V_i}^{V_f} dV$$

$$w = -p_{\text{ex}}(V_f - V_i)$$

For $\Delta V = V_f - V_i$,

$$w = -p_{\text{ex}}\Delta V$$

Free Expansion

$$w = -p_{\text{ex}}\Delta V$$

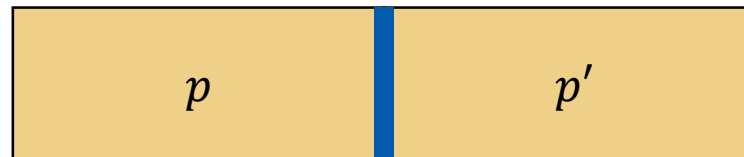
$$\text{If } p_{\text{ex}} = 0, w = 0$$

(regardless of the value of ΔV)

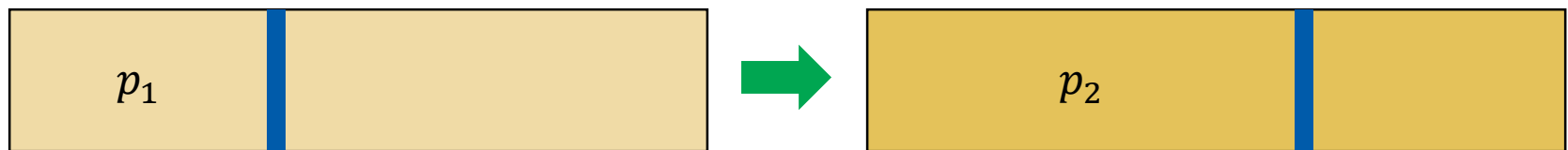
e.g. a gas expanding into a vacuum

Reversible Expansion

- Reversible change: a change that can be reversed by an **infinitesimal** modification of a variable
- The system is **at equilibrium** at any time during the change



$p = p'$ always



Reversible Expansion Work

$$w = - \int_{V_i}^{V_f} p_{\text{ex}} dV$$

For the reversible expansion, $p = p_{\text{ex}}$ always.

$$w = - \int_{V_i}^{V_f} p dV$$

Case of Perfect Gas

$$w = - \int_{V_i}^{V_f} p \, dV$$

For the reversible expansion of a perfect gas,

$$pV = nRT$$

$$p = nRT/V$$

Hence,

$$w = - \int_{V_i}^{V_f} p \, dV = - \int_{V_i}^{V_f} \frac{nRT}{V} \, dV = -nR \int_{V_i}^{V_f} \frac{T}{V} \, dV$$

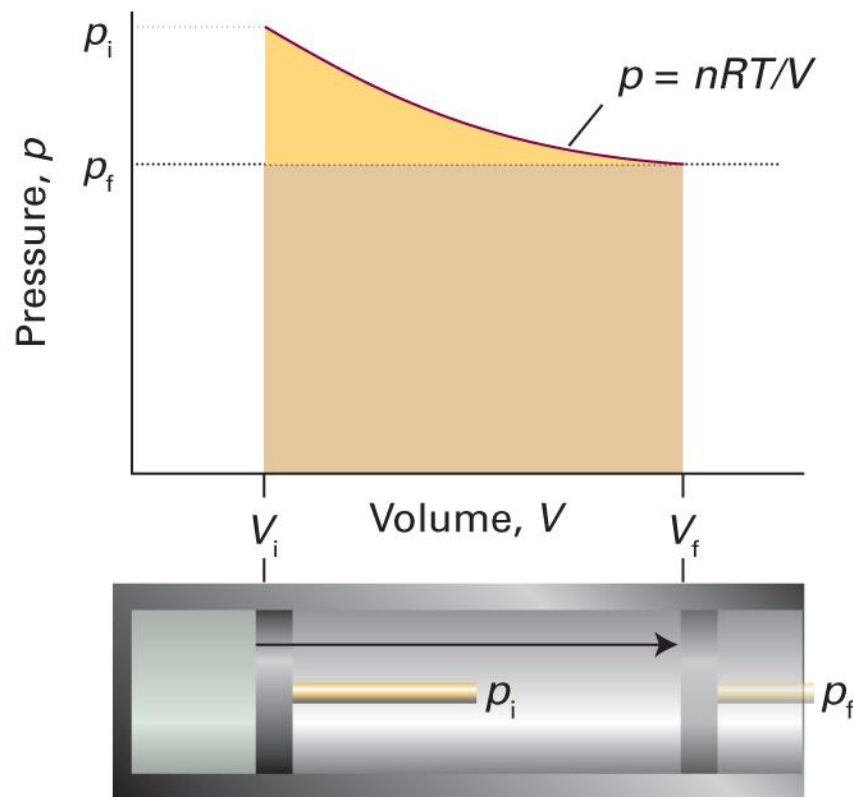
Perfect Gas, Constant T

For an **isothermal** reversible expansion,

$$w = -nR \int_{V_i}^{V_f} \frac{T}{V} dV = -nRT \int_{V_i}^{V_f} \frac{1}{V} dV$$

$$w = -nRT (\ln V_f - \ln V_i) = -nRT \ln \frac{V_f}{V_i}$$

Perfect Gas, Constant T



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 2A.7)

Internal Energy

$$dU = \delta q + \delta w \text{ (1st law)}$$

$$dU = \delta q + \delta w_{\text{exp}} + \delta w_{\text{add}}$$

$$dU = \delta q - p_{\text{ex}}dV + \delta w_{\text{add}}$$

For $\delta w_{\text{add}} = 0$,

$$dU = \delta q - p_{\text{ex}}dV$$